

Sixteen words cover all 15-bit strings

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Abstract

From every binary string of length $n = 3k$ delete k bits, choosing the deletions so that as few distinct strings of length $2k$ remain as possible. The least number $H(3k, k)$ grows like α^{3k} . We give sixteen strings of length 10 such that every string of length 15 contains one of them as a subsequence, so $H(15, 5) \leq 16$, improving the bound 17 from 2013, and hence $\alpha \leq 16^{1/15} < 1.2031$. The set is closed under complement and reversal, and among such sets sixteen is optimal; that claim is a computer result, certified by a checked unsatisfiability proof. The argument from the cover to the bound on α is checked in the Lean proof assistant.

1 The question

Take all 2^n binary strings of length n , with $n = 3k$, and from each delete exactly k bits. The deletions may be chosen separately for each string, and the aim is to leave as few different strings of length $2k$ as possible. How few? The question was asked on MathOverflow in 2013 [2]. Stop and try $n = 3$ before reading on: every string of length 3 has two equal bits, so deleting the odd bit leaves 00 or 11, and two strings suffice.

Write $H(n, b)$ for the least number of strings of length $n - b$ that are left when b bits are deleted from each string of length n . Equivalently, it is the size of the smallest set S of strings of length $n - b$ such that every string of length n contains a member of S as a subsequence; we say S *covers* length n . In coding-theory terms S is a binary insertion covering code [1, 3]; those papers bound such codes for a fixed number of insertions, while here the number grows with the length.

The 2013 discussion found the following. Splitting a string into blocks and treating each block separately gives $H(n_1 + n_2, b_1 + b_2) \leq H(n_1, b_1) H(n_2, b_2)$, so by Fekete's lemma $\alpha = \lim H(3k, k)^{1/(3k)}$ exists and equals the infimum (Meyerowitz, Sawin [2]). Each small cover gives an upper bound: $H(3, 1) = 2$ gives $2^{1/3} \approx 1.260$, Goucher's six words for $n = 9$ give $6^{1/9} \approx 1.220$, Meyerowitz's ten words for $n = 12$ give $10^{1/12} \approx 1.212$, and a seventeen-word cover for $n = 15$, recorded in the question, gives $17^{1/15} \approx 1.208$. Counting, as McKay and Yury showed, gives $\alpha \geq 2^{5/3}/3 \approx 1.058$ [2]. The question also records $12 \leq H(15, 5)$.

Theorem 1. *The sixteen strings of length 10*

0000000000	0000000011	0000111110	0011001100
0011111111	0110000110	0110101001	0111110000
1000001111	1001010110	1001111001	1100000000
1100110011	1111000001	1111111100	1111111111

cover length 15: every binary string of length 15 contains one of them as a subsequence. No proper subset does. Hence $H(15, 5) \leq 16$.

Corollary 1. $\alpha \leq 16^{1/15} < 1.2031$, against $17^{1/15} > 1.2078$ before.

Proposition 1. *No set of fifteen or fewer strings of length 10 that is closed under complement and under reversal covers length 15.*

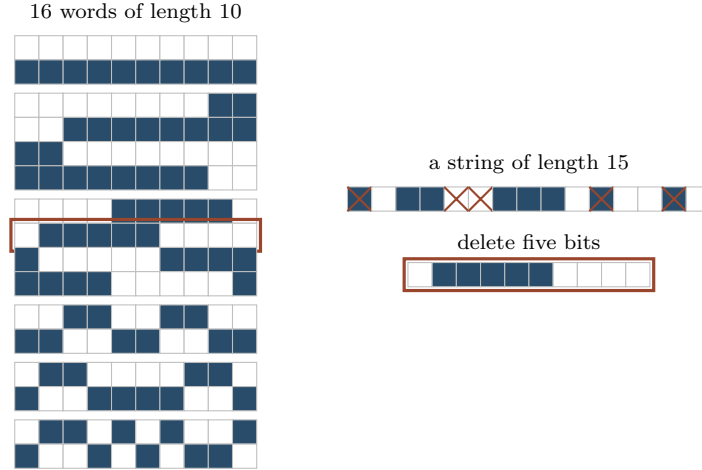


Figure 1: Left: the sixteen words, one per row (dark cells are 1), grouped into the six orbits of complement and reversal. Right: a string of length 15; deleting the five crossed bits leaves the outlined word.

Proposition 1 is a computer result, so it is worth saying how it is checked. The search space is described in Section 3; the infeasibility was found by one solver, confirmed by a second with a different encoding, and the second solver’s proof of unsatisfiability was checked by an independent proof checker. Whether a cover of size 15 exists without the symmetry is open. A 100-second unrestricted search neither found one nor ruled it out.

2 From a cover to the growth rate

Lemma 1 (blocks). *If S covers length n_1 with strings of length m_1 and T covers length n_2 with strings of length m_2 , then the concatenations $\{st : s \in S, t \in T\}$ cover length $n_1 + n_2$ with strings of length $m_1 + m_2$, and there are at most $|S||T|$ of them.*

Proof. Split a string of length $n_1 + n_2$ into its first n_1 bits and the rest. The first part contains some $s \in S$ as a subsequence and the second some $t \in T$, so the whole contains st . $\square$

So H is submultiplicative, $\log H(3k, k)$ is subadditive in k , and Fekete’s lemma gives the limit α as an infimum. In particular $\alpha \leq H(15, 5)^{1/15} \leq 16^{1/15}$, which proves Corollary 1. Concretely, the m -fold concatenations of the sixteen words cover length $15m$ with 16^m strings.

3 How the cover was found

Covering length 15 with strings of length 10 is a set-cover problem with $2^{10} = 1024$ candidate strings and $2^{15} = 32768$ strings to cover. Meyerowitz suggested in the 2013 thread looking for covers closed under complement and reversal [2]. Both maps preserve the subsequence relation (applied to both strings), so a union of orbits covers everything as soon as it covers one string from each orbit of length-15 strings. The 1024 strings fall into orbits of size 2 or 4, and one Boolean variable per orbit gives an exact model: one clause per string of length 15 (or per orbit of them), listing the orbits with a member that is a subsequence of it, and a weighted bound on the total size.

An exact solver (CP-SAT) finds a symmetric cover of size 16 and proves that none of size 15 exists, in a few seconds on one core. The recheck builds its own clauses and its own cardinality

encoding, solves them with a different solver (Glucose), and writes a proof of unsatisfiability for size 15 that the independent checker drat-trim verifies. The cover in Theorem 1 is then checked by itself: a direct test of all 2^{15} strings, in two programs that share no code. Each of the sixteen words is needed: removing any one leaves some string of length 15 uncovered.

Remark. To beat $16^{1/15}$ with the next length, $n = 18$, needs a cover by at most 27 strings of length 12. An eight-minute symmetric search found nothing below 30; that proves nothing either way. The exact value of α , which lies between 1.058 and 1.2031, is open, and so is $H(15, 5)$, which lies between 12 and 16.

Verification. Lemma 1, the submultiplicativity of H , the existence of α by Fekete’s lemma (Mathlib’s `Subadditive.tendsto_lim`), the bound $\alpha \leq 16^{1/15}$ from a sixteen-word cover, and the numerical bounds on $16^{1/15}$ and $17^{1/15}$ are checked in Lean 4 with Mathlib [4], using only the standard axioms. The cover itself and Proposition 1 are finite computations, checked as described in Section 3 and not in Lean. Everything is at <https://github.com/jvkv/mathematics/tree/main/sixteen-words>.

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